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PERFORMANCE SPECIFICATION

CLOTH, WATERPROOF, WEATHER RESISTANT

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for a weather resistant, waterproof, cloth.

1.2 Classification. This specification covers the following types and classes for a weather resistant, waterproof, cloth.

1.2.1 Types.

Type I	-	Cloth, moderate strength, dimensionally stable
Type II	-	Cloth, high strength, dimensional stable
Type III	-	Cloth, ultra-high strength, dimensional stable
Type IV	-	Cloth, high strength

1.2.2 Classes.

Class 1	-	Extreme cold temperature flexibility
Class 2	-	Cold temperature flexibility, flame retardant
Class 3	-	Extreme cold temperature flexibility, flame resistant
Class 4	-	Deleted
Class 5	-	Extreme temperature resistance

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1487 - Glossary of Cloth Coating Imperfections
MIL-STD-3064 – Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

Title 16 CFR Part 303 – Rules and Regulations Under the Textile Fiber Products Identification Act
Title 16 CFR Part 1500 – Federal Hazardous Substances Act Regulations
Title 29 CFR Part 1910 – Occupational Safety and Health Standards
Title 40 CFR - Protection of Environment

(Copies of these documents are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM111 - Test method for Weather Resistance of Textiles: Exposure to Daylight and Weather

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(Copies of these documents are available online at https:/aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D471	- Standard Test Method for Rubber Property – Effect of Liquids
ASTM D751	- Standard Test Methods for Coated Fabrics
ASTM D572	- Standard Test Methods for Rubber – Deterioration by Heat and Oxygen
ASTM D2261	- Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine)
ASTM D3776/D3776M	- Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test methods for Visually Inspecting and Grading Fabrics
ASTM D5035	- Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)
ASTM D6413/D6413M	- Standard Test Method for Flame Resistance of Textile (Vertical Test)

(Copies of these documents are available online at https://www.astm.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J1966 - Lubricating Oil, Aircraft Piston Engine (Non-dispersant Mineral Oil), Standard

SAE-AMS-STD-595 - "Colors Used in Government Procurement"

(Copies of this document are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Standard sample. Unless otherwise indicated, the finished waterproof, weather resistant cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Dusting powder. When applicable, the dusting powder shall be a commercial product whiting, talc, or other finely divided mineral material that does not support mildew growth (4.6.13). Talc shall not contain asbestos and whiting shall contain less than 1.0 percent quartz. The cloth may be dusted on both sides with the dusting powder to prevent blocking.

3.4.2 Cloth performance requirements. The waterproof, weather resistant cloth shall conform to the requirements specified in 3.5 through 3.8, when tested as specified in 4.6. The cloth shall be compatible with commercial state of the art seam sealing materials and technology.

3.5 Color. The waterproof, weather resistant cloth shall be in accordance with SAE-AMS-STD--595 or as specified in the contract or procurement documents (see 6.2).

3.5.1 Visual shade matching. The color and appearance of the dyed waterproof, weather resistant cloth shall match the standard sample, when tested as specified in 4.7.1.

3.6 Mildew resistance. The class 2 cloth shall be mildew resistant. If plasticizers are used, they shall be restricted to phosphate or phthalate esters for the purpose of mildew resistance (see 4.6.13).

3.7 Physical requirements. The finished waterproof, weather resistant cloth shall conform to the requirements, listed below in Table I, when tested as specified in 4.6.

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TABLE I. Physical requirements.

Characteristic	Type I			Type II and IV			Type III
	Class 1	Class 2	Class 3	Class 1	Class 2	Class 3	Class 5
Weight, oz./sq.yd. (± 1.0)	8.0	9.0	10.5	16.0	18.0	18.0	12.0
Breaking strength, lbs (min.)							
Initial:							
Warp	115	115	115	225	225	225	325
Filling	115	115	115	225	225	225	275
After accelerated weathering:							
Warp	92	92	92	180	180	180	260
Filling	92	92	92	180	180	180	220
Tearing strength, lbs (min.)							
Warp	12	12	12	45	45	45	20
Filling	10	10	10	38	38	38	20
Stiffness, cm (max.) (warp only)							
At 70ºF	10.0	10.0	10.0	14.5	16.5	14.5	--
At 10ºF	--	13.0	--	--	22.0	--	--
At -40ºF	12.0	--	12.0	20.0	--	20.0	--
Adhesion of coating, lbs/2-inch width (min.) 1/	10.0	10.0	10.0	15.0	15.0	15.0	12.0
Blocking, rating (max.)	3	3	3	3	3	3	2
Resistance to accelerated aging	2/	--	2/	2/	--	2/	2/
Flame resistance (max.)							
After flame (sec.)	--	--	10	--	--	10	--
Char length (in.)	--	--	3.5	--	--	3.5	--
Flame retardancy	--	3/	--	--	3/	--	
Hydrostatic resistance (min.) Initial and after abrasion	4/	4/	4/	4/	4/	4/	4/
Resistance to:							
Oil	5/	5/	5/	5/	5/	5/	--
Aromatic hydrocarbons	6/	6/	6/	6/	6/	6/	--
Accelerated weathering	7/	7/	7/	7/	7/	7/	7/
Low temperature	--	--	--	--	--	--	6/
High temperature	--	--	--	--	--	--	8/

1/ Requirement applies if a film or coating is applied to the surface of the cloth.

2/ The cloth shall show no softness, stiffness, tackiness or brittleness when compared with a unexposed specimen.

3/ The tip of the flame shall not pass beyond the top edge of the test specimen before 42 seconds after the start of the burner flame.

4/ No leakage in any portion of the test specimen.

5/ No evidence of seepage of oil through the cloth.

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6/ The cloth shall not crack or flake.
7/ The cloth shall show no evidence of cracking, crazing, blooming, chalking or appreciable color change.
8/ The cloth shall show no softness, tackiness or blistering.

3.8 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished waterproof, weather resistance, cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7.14.

3.9 Back side identification. The back side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Back" at the beginning and end of each roll. The marking shall not show through onto the face side of the cloth.

3.10 Width. For Government procurements only, the width of the finished cloth shall be as specified in the contract or procurement documents (see 6.2) and shall be the minimum acceptable width exclusive of the selvage.

3.11 Length and put-up. For Government procurements only, the cloth shall be furnished in lengths as specified in the contract or procurement documents (see 6.2). Each length shall be put-up full width on a roll as specified in 5.1.

3.12 Fiber identification. . Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with Title 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act.

3.13 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4 shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified in this specification or applicable procurements documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.3 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects listed in 4.5 and tested for the characteristics as specified in 4.6 through 4.7.

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4.4 Conformance inspection. Conformance inspection shall include the visual examination of 4.5 and the tests of 4.6 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.5 Visual examination. Each roll in the sample shall be examined, on one side only, for the defects listed in MIL-STD-1487. However, the side of the cloth examined shall be alternated on every other roll examined (except for Type III, Class 5 cloth). Odors of chemicals or commonly used finishes (i.e., films and chemical compounds) shall not be regarded as objectionable. The same yardage shall be given a through-lighting inspection for pinholes and thin areas in accordance with MIL-STD-1487. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warp-wise yard or fraction thereof in which it occurs. The sample unit shall be one linear yard. An approximate equal number of yards shall be examined from each roll in the sample.

4.5.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.5.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.6 End item testing. The cloth shall be tested for the characteristics listed in Table II. The methods of testing as specified wherever applicable and as listed in Table II shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yard full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. Material testing.

Characteristic	Requirement Reference	Test Method
Weight	Table I	ASTM D3776/D3776M, Method C
Break strength: Initial After accelerated weathering	 Table I Table I	 ASTM D5035 (1C-E or 1C-T) 4.7.2, 4.7.3 & ASTM D5035 (1C-E or 1C-T)
Tearing strength	Table I	ASTM D2261, 1/
Stiffness: At 70 (± 2)°F At 10 (± 2)°F At -40 (± 2)°F	 Table I	 See 4.7.4
Adhesion of coating	Table I	ASTM D751 and 4.7.5
Blocking	Table I	ASTM D751 and 4.7.6
Accelerated aging	Table I	ASTM D572 and ASTM D5034 2/
Flame resistance	Table I	ASTM D6413/D6413M
Flame retardancy	Table I	ASTM D6413/D6413M 3/
Hydrostatic resistance Initial After abrasion	 Table I Table I	 ASTM D751 and 4.7.7.1 4.7.7.2 and 4.7.7.1
Oil resistance	Table I	4.7.8
Resistance to: Aromatic hydrocarbon Accelerated weathering Low temperature High temperature	 Table I Table I Table I Table I	 ASTM D471 and 4.7.9 4.7.3 and 4.7.10 4.7.11 4.7.12
Toxicity	3.8	4.7.14

1/ Specimens of Type IV cloth shall be 8-inches by 10-inches.
2/ Only one specimen shall be tested for an exposure .time of 7 days.
3/ Except that the specimen shall be observed for an additional 30 seconds after burner flame is turned off. If the tip of the flame passes beyond the top edge of the specimen at any time during the 42 seconds after the burner is turned on, the specimen shall be reported as a test failure. Five (5) specimens shall be tested in the warp direction and the results reported individually as "pass" or "fail".

4.7 Methods of testing and inspection.

4.7.1 Visual shade matching. The color and appearance of the cloth shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.5) with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

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NOTE: Cloth which has been dusted with dusting powder shall be wiped with a wet cloth and allowed to dry before color matching.

4.7.2 Breaking strength after accelerated weathering. The specimens shall be cut from each test sample unit at the same time as the specimens for initial test and shall be marked to insure proper identification. The specimens shall be subjected to the accelerated weathering procedure specified in 4.7.3. After completion of the procedure, the two specimens shall be subjected to the following conditions:

a. Two (2) hours at 180 (± 5)ºF
b. 40 (± 8) hours at 70 (± 2)ºF at a relative humidity of 65 (± 2)ºF

Each specimen shall then be tested for breaking strength in accordance with ASTM D5035 (1C-E or 1C-T).

4.7.3 Accelerated weathering procedure. One specimen, 8 by 6-inches minimum, with the short dimension warp wise and one specimen, 8 by 6-inches, minimum, with the short dimension filling wise shall be subjected to the procedure specified in AATCC TM111, Option A. The procedure shall be performed with the face side exposed. The exposure times shall be as follows:

Type I, II and IV, Class 2	– 150 hours for all colors.
Type I, II and IV, Class 1 and 3	– 150 hours for Olive Green and black only. All other colors – 50 hours.
Type III, Class 5	– 150 hours for all colors

After exposure, the specimens shall be removed and allowed to condition for at least 24 hours.

NOTE: Equipment which operates at one revolution in two hours may be used. However, the one revolution per minute shall be the standard procedure in the event of dispute. Laboratory reports shall state the procedure used.

4.7.4 Stiffness test. Stiffness testing shall be in accordance with the following procedure, except that the apparatus and test specimens shall be conditioned for a minimum of four (4) hours before testing and the test shall be performed in a still atmosphere at specified temperature.

Procedure.

- Five (5) specimens from the warp direction only shall be cut measuring 1 1/4-inches by 6 to 12-inches with the longer dimension parallel to the direction of the warp yarns being tested. Testing shall be performed at 70ºF, 0ºF and -40ºF using a Clark Paper Softness-Stiffness tester.

- The measure of directional flex-stiffness shall be the length of the specimen remaining above the nip of the rolls which, when the test instrument is rotated clockwise and counter-

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clockwise through 90 (±2) degrees, just falls to the left and right of the line perpendicular to the nip of the rolls. It is this length of specimen which yields the flexibility criteria being measured. The specimen shall be considered to have fallen to the right or left when the specimen, during the course of clockwise and counter-clockwise rotation, passes across the line perpendicular to the nip of the rolls as indicated by the pointer fastened to the rotating framework.

The length of the specimen shall be determined as follows:

Initial adjustment of test instrument.

1. The apparatus shall be placed in the test position and leveled with the axis of the clamping surface (nip of the rolls) horizontal and the pointer, fastened to the rotating framework, in the vertical position.

2. One end of the specimen shall be inserted between the rolls with the lengthwise edges of the specimen perpendicular to the nip of the rolls.

3. The specimen shall be positioned and shall be of sufficient length to allow it to bend to the left of the line perpendicular to the nip of the rolls, which when the apparatus is rotated, with continuous uniform motion in the clockwise direction it will fall to the right of the perpendicular line.

4. At the point the specimen falls to the right, the instrument shall be stopped and the adjustable circular scale moved to make the zero position on the scale coincide with the pointer fastened to the rotating framework. The instrument shall then be rotated with continuous uniform motion (±2) degrees from this point in the counter-clockwise direction. The specimen should not fall to the left of the perpendicular line at this point.

5. If the specimen falls to the left then the instrument shall be rotated in the clockwise direction until the pointer is vertical. The specimen shall be shortened and reset with a bend to the left and the above procedure in 4., shall be repeated.

Determination of flex-stiffness.

1. After initial adjustment the instrument is rotated until the pointer is vertical. The length of the specimen is shortened by a small increment, set with a bend to the left and the instrument is rotated with continuous uniform motion clockwise until the specimen falls to the right to the perpendicular line. The instrument is stopped and the zero position of the adjustable scale is made to coincide with the pointer. The instrument is rotated, with continuous uniform motion, in the counter-clockwise direction 90 (±2) degrees when the instrument is immediately reversed in the clockwise direction and rotated, with continuous uniform motion, to return the pointer to zero. The specimen shall have fallen immediately to the left of perpendicular at 90 (±2) degrees and shall fall immediately to the right of the perpendicular as the pointer reaches zero.

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2. If the specimen does not fall to the left when it is rotated the required 90 (±2) degrees in the counter-clockwise direction the pointer is rotated to the vertical position, the specimen is shortened another small increment and the procedure in 1., shall be repeated.

3. When the specimen has reached the length such that it falls to the left and right of the line perpendicular to the nip of the rolls the instrument is stopped with the pointer in a vertical position.

4. Without tension the specimen above the rolls shall be extended and the length measured from the nip of the rolls to the end of the extended specimen, to the nearest mm.

5. The length of the extended specimen is recorded as the directional flex-stiffness of the specimen.

Report of results.

The directional flex-stiffness of the sample unit tested shall be reported as the average of the specimens tested in the warp direction. Individual values used to calculate the average shall also be reported.

4.7.5 Adhesion of coating. In accordance with ASTM D751, Test method, Adhesion of Coating with 2-inch wide reinforced coating adhesion specimens, cyanoacrylate (solventless) adhesive, and pulling clamp speed of 5 mm/s. The test shall be performed on the coated side (if one side coated) or face to back (if two sides coated) of the cloth. Testing shall be performed in the warp direction.

4.7.6 Blocking. In accordance with ASTM D751, Test method - Determination of Blocking Resistance of Fabrics Coated with Rubber or Plastics at Elevated Temperatures, except that the specimens shall be exposed at an oven temperature of 180 (±2)ºF for 30 minutes. Resistance to blocking shall be rated in accordance with the following scale:

1 – *No Blocking*. Cloth surfaces are free and separate without any evidence of cohesion or adhesion.
2 – *Trace Blocking*. Cloth surfaces show slight cohesion or adhesion.
3 – *Slight Blocking*. Cloth surfaces must be lightly peeled to separate.
4 – *Blocking*. Cloth surfaces separate with difficulty or coating is removed during separation.

4.7.7 Hydrostatic resistance.

4.7.7.1 In accordance with ASTM D751, Test method - Hydrostatic Resistance, Procedure B (Rising Water Column Tester), Procedure 2 with the hydrostatic head fixed at 50 centimeters and applied to the test specimen for 10 minutes. The back side of the cloth shall contact the water. The report shall only include “measurement of the appearance of water drops”. Leakage of any specimen shall be considered a test failure. Leakage is defined as the appearance of water at three or more different places within the 4 1/2-inch diameter test area.

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4.7.7.2 A new specimen, 10 by 10 inches, shall be abraded by means of a 2-inch square piece of grade 1/0 garnet paper which shall be uniformly loaded with an 8.0 (± 0.)1 ounce weight. The specimen shall be abraded on the face by moving the weighted garnet paper filling-wise five (5) times in each direction. The specimen shall then be turned over and abraded on the back by moving the weighted garnet paper warp-wise five (5) times in each direction. The abraded 2-inch square area shall then be placed face up across the center line of the clamping head so that the center of the abraded area will coincide with the center of the exposed part of the specimen. Hydrostatic resistance shall then be determined as specified in 4.7.7.1.

4.7.8 Oil resistance. One (1) 8 by 8-inch specimen shall be placed on a wood frame. The inside dimensions of the frame shall be 6 by 6-inches by 1-inch. The specimen shall be forced into the frame by a wood block 5 3/4 by 5 3/4-inches (with round corner) to form a basin of uniform depth. The edges of the cloth shall be tacked to the frame and the block removed. Lubricating oil, conforming to Grade 50 of SAE J1966 shall be rapidly poured into the basin to a 1/2-inch depth. After the oil has been in the basin for one (1) hour, the bottom of the fabric specimen forming the basin shall be examined to determine if oil has seeped through or permeated the cloth.

4.7.9 Resistance to aromatic hydrocarbon fluid. One (1), 1 by 6-inch specimen with the long dimension parallel to the warp shall be immersed for five (5) minutes in aromatic hydrocarbon fluid in accordance with Reference fuel D of ASTM D471. The specimen shall be dried at room temperature for two (2) hours (± 5) minutes and then be creased sharply on itself, with the face side toward the outside of the fold. A similar specimen, except with the long dimension parallel to the filling shall be tested in the same manner.

4.7.10 Resistance to accelerated weathering. The cloth shall be subjected to the accelerated weathering procedure specified in 4.7.3. After completion of the procedure, each specimen shall be folded by hand, face out, sharply upon itself and visually examined along the fold for cracking and crazing. Each specimen shall then be opened flat and examined for blooming, chalking or appreciable color change. (An appreciable color change is one that is noticeable on first glance when comparing the tested specimen with the original unexposed cloth).

4.7.11 Resistance to low temperature. A 1 by 4-inch specimen of the cloth with the long dimension warp-wise and a 1 by 4-inch specimen of the cloth with the long dimension filling-wise shall be exposed for 4 hours at a temperature of minus 70 (± 5)°F with the temperature recorded at the lowest point in the chamber. The sample, still in the test atmosphere, shall then be sent sharply, face side of the cloth to the fold, 180 degrees over a 1/8-inch steel rod that has been exposed in the test chamber with the test specimen.

4.7.12 Resistance to high temperature. A 2 by 6-inch specimen of the cloth shall be exposed for a period of 6 hours in an electrically heated oven maintained at a temperature of 170 (± 2)°F. At the end of the exposure period, the specimen shall, while still maintained at the test temperature, be bent 180 degrees over a 1/8-inch steel rod with the face side of the cloth to the outside of the fold.

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4.7.13 Mildew resistance. The contractor shall provide scientific data analysis or Material Safety Data Sheets (MSDS) that stated requirement for mildew resistance has been met.

4.7.14 Toxicity test. When the toxicity requirement (see 3.8) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (see 6.4 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I, II and IV cloths are for use in the fabrication of covers and shelters. Type III cloth is intended for use in the manufacturing of survival containers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type and class required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1, 4.3)
e. Conformance inspection acceptance quality limits (AQLs) (see 4.4)
f. Color and standard sample required (see 3.5)
g. When toxicity testing is required (see 3.8, 4.7.14)
h. Width of cloth required (see 3.10)
i. Length required if other than specified (see 3.11)
j. Inspection conditions (see 4.2)
k. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

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6.5 Visual shade matching. In 2017, Option A of AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

6.6 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1)
Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100)
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250) **

6.7 Subject term (key word) listing.

Fabric
Coated
Mildew resistant
Oil

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA-CT

(Project: 8305-2022-021)

Review activities
Army – MD, AV, MI, AT
Navy – MC

NOTE: The activities listed above were interested in this document as of date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:13Z
Check the source to verify that this is the current version before use.